

Statistics Practice Set — S20

S20 moves one step beyond S19. The focus is now **multi-step analysis, interpreting conflicting statistics, and choosing the right measure for a business decision.**

No basic Population/Sample question. No formula-recall questions.

Q1 — Mean vs Median: Revenue Analysis

A shop records the following daily revenue:

Daily Revenue
₹800
₹850
₹900
₹950
₹1,000
₹1,100
₹1,200
₹8,000

Answer:

1. Calculate the mean.
 2. Calculate the median.
 3. Which measure better represents a typical day?
 4. Is the dataset likely right-skewed or left-skewed?
 5. Explain the effect of ₹8,000 on the mean.
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Q2 — SD vs CV: Two Production Lines

Production Line	Mean Output	SD
A	500 units	40
B	2,000 units	120

Answer:

1. Calculate CV for A and B.
 2. Which line has greater absolute variability?
 3. Which line has greater relative variability?
 4. Which line is more consistent relative to its output?
 5. Why would comparing only SD be misleading?
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Q3 — IQR Resistance to Outliers

Two datasets contain employee overtime hours.

Dataset A

10, 12, 14, 15, 16, 18, 20

Dataset B

10, 12, 14, 15, 16, 18, 80

Answer:

1. Calculate the range of A and B.
2. Calculate Q1 for both.
3. Calculate Q3 for both.
4. Calculate IQR for both.
5. Which measure is more resistant to the extreme value?
6. What does this tell you about using range for business analysis?

Q4 — Five Number Summary + Shape

A dataset has:

Statistic	Value
Minimum	12
Q1	20
Median	28
Q3	35
Maximum	80

Answer:

1. Find the lower whisker length.
2. Find the upper whisker length.
3. Which side has the longer tail?
4. Identify the likely skewness.
5. Give **two pieces of evidence** supporting your conclusion.

Q5 — Box Plot Comparison

Two departments have the following salary distributions:

Statistic	Department A	Department B
Minimum	₹20,000	₹15,000
Q1	₹30,000	₹25,000
Median	₹40,000	₹38,000
Q3	₹48,000	₹60,000
Maximum	₹55,000	₹1,00,000

Answer:

1. Which department has the higher median?
 2. Calculate IQR for both.
 3. Calculate range for both.
 4. Which department has greater variability?
 5. Which department appears more consistent?
 6. Which department has the longer upper tail?
 7. Which department appears more likely to have extreme high salaries?
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Q6 — Outlier Detection + Decision

A company's delivery times are:

11, 13, 15, 16, 18, 19, 21, 23, 25, 60

Using the IQR method:

1. Calculate Q1.
 2. Calculate Q3.
 3. Calculate IQR.
 4. Calculate the lower fence.
 5. Calculate the upper fence.
 6. Identify the outlier.
 7. Should the company use mean or median to report the typical delivery time?
 8. Explain your choice.
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Q7 — CV Ranking

Three investment options have:

Investment	Mean Return	SD
A	12%	2.4%
B	20%	4%
C	30%	4.5%

Answer:

1. Calculate CV for all three.
 2. Rank them from **most consistent** → **least consistent**.
 3. Which investment has the highest return?
 4. Which has the lowest relative variability?
 5. Does the investment with the highest return also have the lowest CV?
 6. If the investor cares about relative consistency, which investment would you recommend?
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Q8 — Conflicting Statistics

Two stores have:

Store	Mean Sales	Median Sales	SD
A	₹50,000	₹49,000	₹4,000
B	₹70,000	₹45,000	₹5,000

Answer:

1. Which store has the higher mean?
2. Which store has the higher median?
3. Which store shows stronger evidence of right-skewness?
4. Which store has greater absolute variability?

5. Which statistic would you trust more for the **typical sale**?
 6. Explain why the mean alone could lead to a misleading conclusion.
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Q9 — Choose the Best Measure

Choose the **best statistical tool** for each situation.

Options:

Mean, Median, SD, IQR, CV, Box Plot

A.

A company wants to report the typical house price in an area containing several luxury mansions.

B.

A manager wants to compare variability between two branches whose average sales are very different.

C.

An analyst wants to measure the spread of the central 50% of observations.

D.

An analyst wants to compare the distribution and possible outliers across eight departments.

E.

Two machines produce the same average number of units, and management wants to compare absolute variation.

F.

A dataset is fairly symmetric with no significant extreme values, and management wants the arithmetic average.

Q10 — Full Data Analyst Case

Two delivery platforms have:

Platform	Mean Delivery	Median Delivery	SD
A	35 min	28 min	7 min
B	50 min	48 min	5 min

Answer:

1. Calculate CV for both platforms.
2. Which platform has higher absolute variability?
3. Which platform has higher relative variability?
4. Which platform shows stronger evidence of right-skewness?
5. Which platform has the more typical/representative mean?
6. Which platform is more consistent relative to its average delivery time?
7. If customers care about **shorter typical delivery time**, which platform is better?
8. If management cares about **relative consistency**, which platform is better?
9. Explain why looking at **mean + median + SD + CV together** gives a better decision than using only one measure.